

CHAPTER 10

Architecture Roles, Skills, and Competencies



This chapter is part of the GREEN and GOLD reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

The previous chapter discussed establishing the Architecture Capability. This activity is one of the first steps to be performed in the Preliminary Phase, along with the formation of an architecture team. Both activities are highly dependent on the ability to define the correct roles, skills, and competencies required to successfully establish the Architecture Capability and associated architecture team. A skills framework can be used to identify these roles and skills.

This chapter focuses on the architecture roles, skills, and competencies provided by the TOGAF® Standard. The framework includes an overview of the roles used in an architecture practice and the skills associated with them. With the release of the “Architecture Roles and Skills” Series Guide, the framework now provides a more comprehensive overview of the skills associated with each role, highlighting the depth of knowledge required to successfully perform the role.

In practice, however, there still seems to be some ambiguity about job descriptions, roles, and assignments. In some situations, this leads to misunderstandings about the roles and responsibilities of architects.

10.1. Difference Between Jobs, Roles, and Assignments

To properly understand the difference between *jobs*, *roles*, and *assignments*, it is essential to clarify these terms, as they can be used in multiple, sometimes overlapping contexts. The Series Guide “Architecture Roles and Skills” defines them as follows:

- **Job:** The *formal position* someone holds, usually defined by a contract of employment. Usually the position is related to the hierarchical positioning in the organization.
- **Role:** The expected *set of activities*, typically tied to a *specific skill set*.
- **Assignment:** A *specific task* or project a person is *responsible for*, often within the scope of their job and role, though not exclusively.

Although the TOGAF Standard makes an excellent attempt to clarify the distinction between jobs, roles, and assignments, there is still little consensus about architecture roles in day-to-day practice. For example, in many organizations today, it is still unclear whether the term *Enterprise Architect* refers to a job or a role. Partly because of this continuing ambiguity, organizations are not well equipped to properly position the profession of architecture within the contours of their own organization. As a result, architecture as a discipline is still invariably viewed as part

of the IT department. Being part of the IT department requires different skills and competencies than being part of a strategic team that translates business strategy into execution.

Much of the misunderstanding has to do with the naming of some architecture roles and their underlying meaning in terms of duties and responsibilities. Most ambiguity occurs when talking about architecture roles in relation to architecture domains. Take the term *Technical Architect*, for instance. Is it a job or a role? And what about an Application or Business Architect? What are their responsibilities? In an effort to eliminate confusion about jobs and roles in architecture domains, the TOGAF Standard created the Segment Architect profile. This profile describes a job, not a role or an assignment. It comes as a complete package, including a role description and assigned responsibilities.

10.2. Core Architecture Roles

The Series Guide “Architecture Roles and Skills” acknowledges that in many organizations, individuals may fulfill multiple roles, while certain roles might also be assigned to multiple individuals. The Series Guide therefore provides a practical *starting point*, showcasing the core roles essential for an Enterprise Architecture practice, which can then be tailored to meet the unique demands of the organization.

The TOGAF Standard outlines an example set of roles critical for supporting the development and governance of an organization’s Architecture Landscape. These roles facilitate robust decision-making across various levels of influence.

The “Architecture Roles and Skills” Series Guide introduces three primary viewpoints: *enterprise*, *segment*, and *solution*. Figure 10-1 presents these viewpoints alongside additional perspectives, explained further below, that enhance this structure.

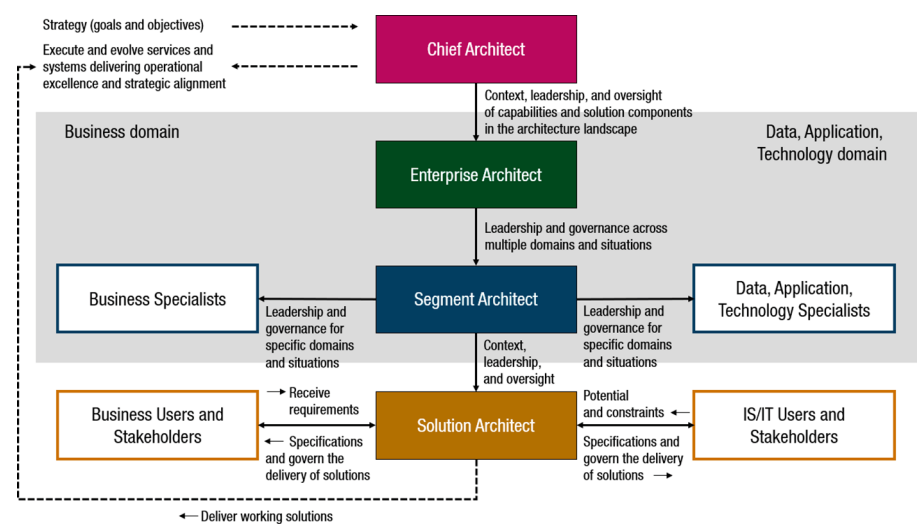


Figure 10-1. Core architecture roles and their relationships

The three primary viewpoints – enterprise, segment, and solution – are equal to the three core architecture roles. The TOGAF Standard does not give specific details on the *specialist*, *user*, and *stakeholder* roles. These can be found in the many different professional frameworks and associations that exist for those roles.

Figure 10-1 shows the *three core architecture roles* – supplemented by the specialized position of Chief Architect – arranged vertically to illustrate the breadth of the roles and to emphasize their interrelationship.

Starting from the Chief Architect at the top and moving down through Enterprise Architect to Segment Architect, each role brings increasing depth in expertise, leadership, and governance, tailored to distinct elements within the Architecture Landscape. This tiered approach supports balanced, well-informed decision-making. Most organizations operate with three to seven levels of management and control, and while many aim to streamline these, the TOGAF Standard’s model starts with a foundational three-tier framework. This minimal setup allows flexibility

to expand as needed for organizations that require additional layers of guidance and governance.

Figure 10-2 omits the additional components and relationships shown in Figure 10-1 in order to properly focus on the three primary architecture roles (with the additional specialization of the Enterprise Architect role, the Chief Architect). The dashed line around the Chief Architect and Enterprise Architect roles in Figure 10-2 is to emphasize their relationship in the sense that the Chief Architect role is a *specialization* of the Enterprise Architect role. They are *not* two separate roles like Segment Architect and Solution Architect.

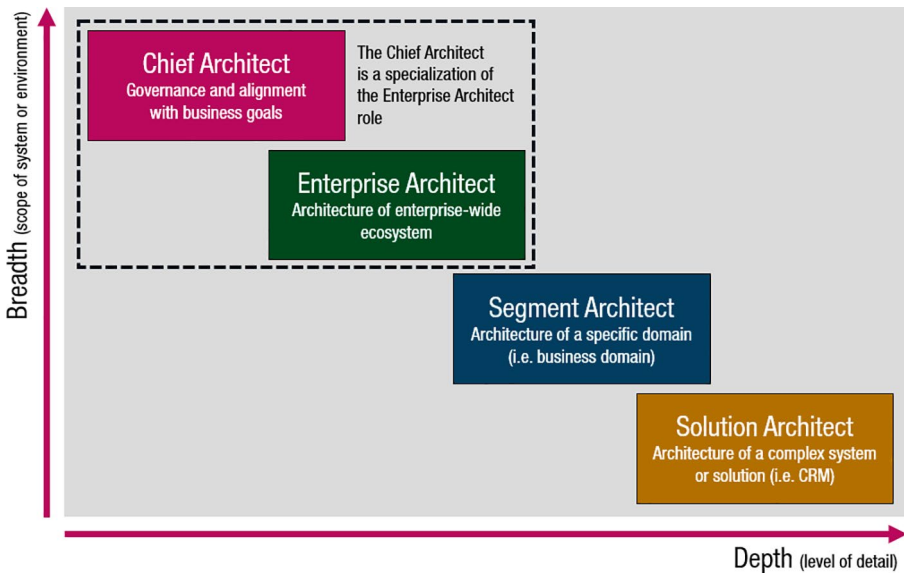


Figure 10-2. Core architecture roles

Chief Architect: The Chief Architect is a special case of an Enterprise Architect and is responsible for integrating the Enterprise Architecture view into the wider strategic and operational context of the organization. This role is performed by a seasoned, influential professional who plays a pivotal role in driving major technical and business transformations across

an organization and even the broader market. Whether an internal leader or a trusted external advisor, they're responsible for building alignment and commitment at all levels, securing buy-in for impactful changes that elevate performance.

Working at the highest levels, the Chief Architect unifies executives and board members around the vision for a robust, forward-looking Architecture Landscape. They make complex technical and business concepts accessible and relevant to leadership, ensuring ideas connect seamlessly with the company's strategic and operational goals.

This role thrives in situations that are dynamic, evolving, or complex. The Chief Architect leads the charge in establishing roadmaps that guide transformative change across various business and technology domains. They shape the Architecture Landscape's structure and direction and take on accountability for its governance and alignment with business goals.

Above all, their mission is to create an environment where strategic change isn't just possible, it's sustainable and impactful. They lead teams of Enterprise, Segment, and Solution Architects, along with specialized experts, to build and manage the services and solutions essential to keeping the organization agile and competitive in a shifting marketplace.

In Europe, few organizations recognize the role of the Chief Architect.

Therefore, the duties and responsibilities associated with this role – as described in the TOGAF Standard – are most often assigned to the Enterprise Architect.

The roles of Enterprise and Chief Architect are typically combined into one.

Enterprise Architect: Seasoned experts, steering transformative change that bridges technology and business across the entire organization, ultimately bringing innovation into the market.

They don't just participate, they *lead* major initiatives, holding a seat at the table with senior decision-makers. With accountability for comprehensive business service delivery through finely tuned information systems, they demonstrate advanced expertise in both technical and business arenas. Their track record showcases substantial achievements and strong, trusted relationships with key stakeholders. Not only do they deliver, but they actively coach and mentor, setting the standard for excellence.

Enterprise Architects thrive in dynamic, complex environments where solutions are uncharted and the scope is fluid. Whether tackling evolving requirements or varied markets or integrating multiple technologies, they build roadmaps that span entire business segments. They define the guiding principles and policies that shape the architecture, ensuring cohesive governance and alignment across all initiatives.

Their core mission is to establish and enhance the framework for integration, interoperability, and governance across solution components. Leading Segment Architects, Solution Architects, and a spectrum of specialists, they shape, deliver, and sustain the solutions that keep their organization competitive in an ever-evolving market.

Segment Architect: Skilled and seasoned professionals who manage and champion the specific business or technical capabilities within a segment of the organization or the broader marketplace.

Their role is all about defining, shaping, and governing the systems, subsystems, and components that bring products and services to life. Segment Architects don't just build roadmaps, they create blueprints for the future of their segments, ensuring every technical and business option is thoroughly evaluated and effectively utilized. They've got policies and standards covered, guaranteeing every element in their segment is effective, efficient, and compliant. Collaboration is second nature for them as they work closely with other architects to ensure seamless connectivity across the entire environment.

Segment Architects excel in dynamic, evolving, or unprecedented scenarios, stepping up to tackle complex, wide-reaching solutions that keep pace with change. They oversee every development effort within their segments, ensuring that all systems and components align with purpose, comply with regulations, and follow established policies and standards.

Their mission is to create the foundation for cohesive integration, smooth interoperability, and continuous evolution across their segments. By setting up clear roles and fostering effective collaboration among technical specialists and Solution Architects, they ensure everyone works in sync to deliver components and building blocks that meet end-to-end business requirements for the organization and its clients.

Solution Architect: Also seasoned professionals who drive the delivery of new or evolved capabilities across the organization and its marketplace.

Their role is to craft and guide the solutions that align with both immediate requirements and broader organizational goals, achieving defined service levels within ambitious timelines. They break down these solutions into systems, subsystems, and components that interact seamlessly and then integrate them into a cohesive roadmap, transforming complex ideas into actionable steps through targeted Transition Architectures.

With an eye for innovation, Solution Architects make sure that each decision leverages the latest business and technical advancements. They are particularly effective in dynamic and complex situations, where solutions must be adaptable to evolving challenges. From specification through to implementation, they ensure each new solution adheres to key principles, policies, and standards, resulting in systems that are not only effective but also efficient and compliant.

Their main mission is to build a collaborative environment that supports seamless solution development, integration, and interoperability. By engaging closely with technical and business teams alike, Solution Architects clarify roles, align expectations, and streamline the delivery of each component, ultimately ensuring the end-to-end service levels are not

just met but exceeded. This collaborative approach allows them to fully understand and address the diverse requirements and expectations of every stakeholder involved.

10.3. The Role of the Segment Architect

Let's take a closer look at the Segment Architect. The person who holds this *job* may very well fill the *role* of Technical Architect (based on the tasks described in the job's *assignment* and the required skills and competencies defined in the role's description). The hierarchical position within the organization is not determined by the job description, but rather by the job assignment. For example, if the job assignment states that the Segment Architect deals with technical infrastructure issues, the role description points to the Technical Architecture domain and requires certain skills and competencies that are useful when working in that particular domain. It's safe to say that this Segment Architect will most likely be part of the IT department.

Conversely, if the Segment Architect's job description states that they are primarily concerned with business issues, then all indications are that the Segment Architect works in the area of Business Architecture. The role description of the second Segment Architect is different from that of a Segment Architect working in the Technical Architecture domain. The skills and competencies required for the two roles are also different. However, both examples involve the same job description, that of a Segment Architect.

The use of *segments* also requires further explanation. This is mainly because there are two ways to look at and use segments and thus to explain the role of the Segment Architect.

The first way to look at segments is from the viewpoint or *perspective of the profession*. This is consistent with how the TOGAF Standard has articulated it. From this perspective (the *horizontal* perspective),

segments, and the role of the Segment Architect, are based on the four architecture domains (Business, Data, Application, and Technology). Each of the domains represents a specialization for the Segment Architect. The specializations are Business Architect, Data Architect, Application Architect, and Technology Architect, corresponding to the four architecture domains.

A *segment* groups capabilities in a structured way, often as part of a larger portfolio or specific change program [19].

The TOGAF Standard also introduces domains, which act as key groupings of capabilities that an organization needs to manage.

A *domain* is used to reflect any useful grouping of capabilities to be managed by an organization, with four particular segment-based viewpoints. These four segments are called the business, application, data, and technology domains.

The second way to look at segments and the role of the Segment Architect is from an *organizational perspective*. This perspective (the *vertical* perspective) is not covered in the TOGAF Fundamental Content and is somewhat hidden in the “Architecture Roles and Skills” Series Guide.

In addition to the usual reference to the four architecture domains, a Segment Architect's job may also be linked to an organizational classification. This classification occurs when business operations are organized into business domains or segments (vertical pillars) rather than using a direct architecture domain classification. When architecture in an organization is considered in terms of business domains or segments instead of architecture domains, it is referred to as an *organizational* or *vertical perspective*. Figure 10-3 illustrates business domains or segments (shown vertically) compared with architecture domains (shown horizontally).

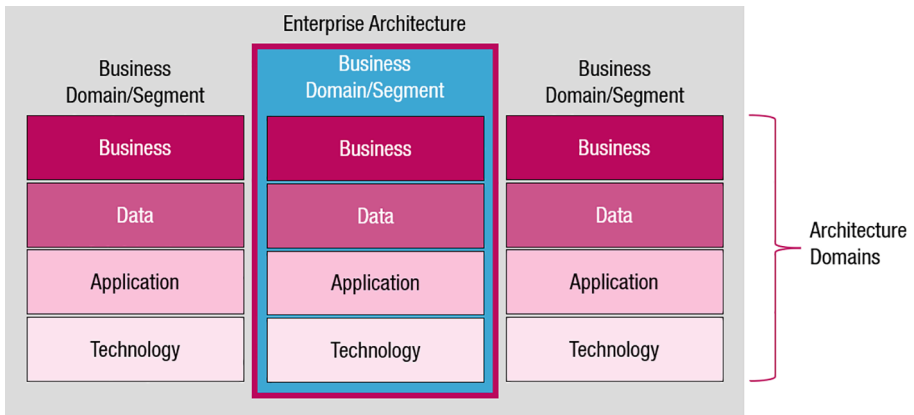


Figure 10-3. *Business domains/segments versus architecture domains*

This vertical perspective is adopted by many organizations, but it makes it difficult to assign architecture roles.

This is because each pillar in the organization actually contains multiple architecture domains. Often, Business Architecture, Data Architecture, and Application Architecture come together in one pillar. The Technology Architecture is typically separated from the other three architecture domains because, in nine out of ten cases, it is the responsibility of IT.

The application of the vertical perspective does not always support the assignment of architecture roles in the way suggested by the TOGAF Standard. Because there are so many different architecture domains in a single pillar – and usually too few architects with the necessary knowledge and expertise in these domains – pragmatic solutions are sought. Typically, this results in an organization creating new architecture roles to meet demand.

For example, I experienced an organization that was working in an agile way. The organization was divided into tribes, and one of the tribes was dealing with architecture issues within one of the pillars of the organization. Since there was no architect who could handle Business Architecture, Data Architecture, and Application Architecture at the same time, the organization created the role of a Tribe Architect. The primary task of this role was to develop architecture within the context of an organizational pillar consisting of Business Architecture, Data Architecture, and Application Architecture at an operational level.

This is just one of many examples of how the choice of organizational perspective often leads to an uncontrollable proliferation of new – and often nonsensical – architecture roles.

The horizontal perspective (based on the architecture domains) and the vertical perspective (which follows business domains or segments) are explained in the following paragraphs.

10.3.1. The Horizontal Perspective

As mentioned in Section 10.3, when viewed from the angle of the architecture profession, the TOGAF Standard recognizes three core architecture roles (with the Chief Architect being a specialization of the Enterprise Architect role). Each of the three core architecture roles is relatively easy to derive from the four Architecture Capabilities mentioned in Chapter 9. Figure 10-4 shows the four Architecture Capabilities, with the corresponding core architecture roles beneath them.

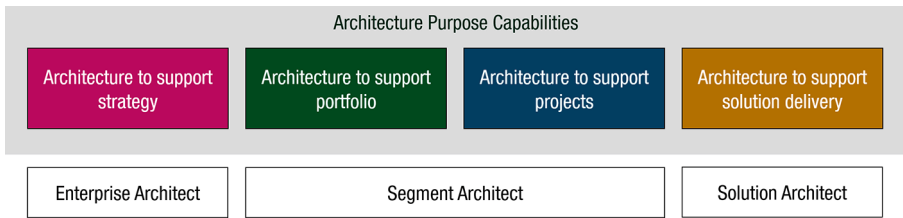


Figure 10-4. *Architecture Capabilities and architecture roles*

The Architecture Capability *in support of strategy* is directly related to the roles of the Chief Architect and the Enterprise Architect. It should be noted that when an organization has a Chief Architect, that role often takes on the strategic side of the discipline. The Enterprise Architect then takes on the development of the architecture for the enterprise ecosystem. If an organization is not large enough to support both roles, then often only the Enterprise Architect exists. In this scenario, the Enterprise Architect will also take on the strategic part of the Architecture Capability.

The two capabilities in the center of Figure 10-4, *in support of portfolio* and *in support of projects*, can be related to the role of the Segment Architect. This role falls into a number of specializations corresponding to the number of architecture domains: Business Architecture, Data Architecture, Application Architecture, and Technology Architecture.

Where the capability supports *solution delivery*, the role of the Solution Architect can be distinguished. Again, like the Segment Architect role, this role has multiple manifestations in the form of specializations (think Cloud, Network, and Software Architect).

Some of the specializations mentioned above that can be associated with the architecture roles by capability are shown in Figure 10-5. Each capability corresponds to an architecture role, and each architecture role corresponds to two or more specializations.

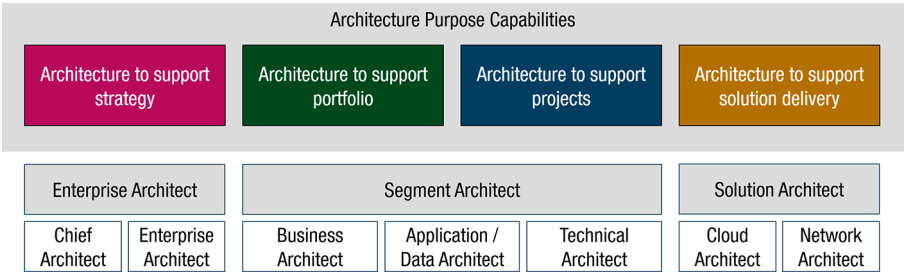


Figure 10-5. *Architecture Capabilities, roles, and specializations*

Figure 10-5 assumes a horizontal perspective in which architecture roles are deployed according to architecture domains. This layout fits well with the four Architecture Capabilities mentioned earlier.

Different architecture roles are required to properly implement each of the four capabilities. When architecture roles are combined into a single role, it creates ambiguity about the deployment of the capabilities and the responsibilities for their implementation and execution.

10.3.2. The Vertical Perspective

In addition to the horizontal perspective (based on the four architecture domains) of looking at architecture roles, the TOGAF Standard also describes a second way of viewing the various architecture roles: from an *organizational perspective*. This perspective is also called the vertical design of an organization. It focuses not on the four architecture domains, but on the *core activities or services* of an organization. The classification that follows is based on organizational pillars per core activity or service. For example, in the healthcare industry, this might result in the pillars of Research, Consultation, and Treatment. Each pillar then consists of a piece of Business Architecture, Data and Application Architecture, and Technology Architecture. In the context of the vertical perspective, a

pillar is also called a segment and thus consists of a Segment Architecture (which is composed of the four domains mentioned earlier). Figure 10-6 illustrates the example just given.

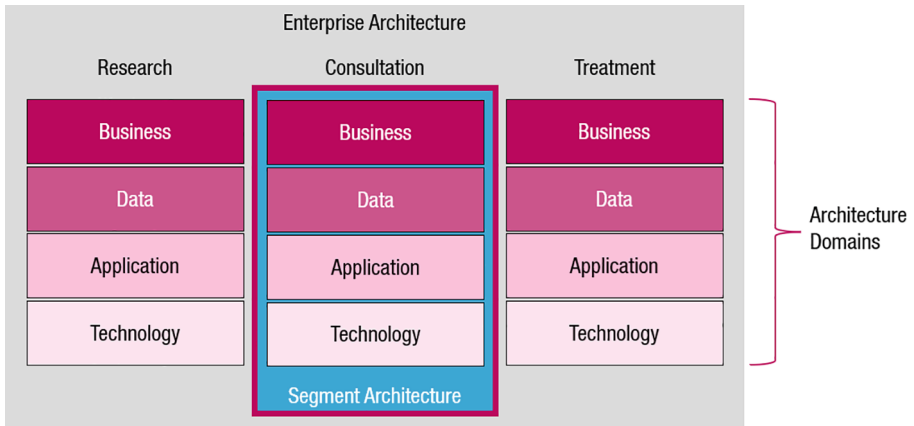


Figure 10-6. *Vertical perspective and associated architectures*

A Segment Architect is responsible for the architecture within an organizational domain (pillar), a segment of the overall Enterprise Architecture. Here, a segment is a vertical domain from an architectural perspective because it encompasses all architectural domains, but specifically from the context of an organization. This is illustrated schematically by the blue block in Figure 10-6.

When an organization adopts organizational segmentation, the work of a Segment Architect often becomes a bit more substantive and operational in nature because of the responsibility for the Application and Technology Architecture domains. This is often the domain of the Solution Architect, which explains the more operational nature of the role.

The difference between a Segment Architecture and a Solution Architecture is illustrated in Figure 10-7. For example, Consultation might be centralized, but its services are used and tasks performed in other processes – used by other departments – as well. This is highlighted by the purple lines in Figure 10-7.

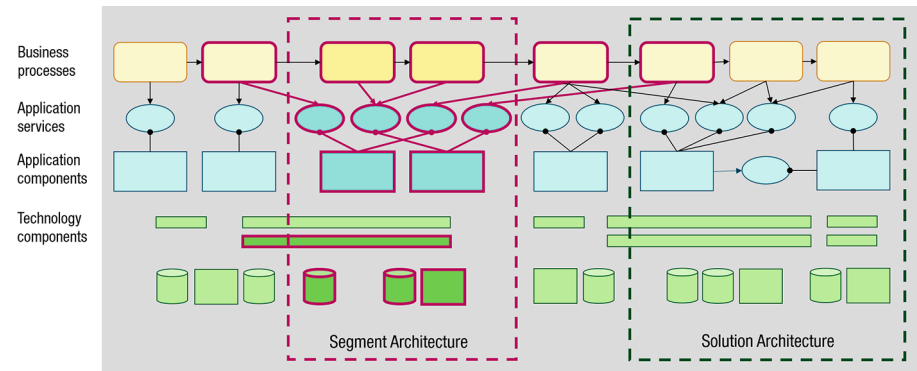


Figure 10-7. Differences between Segment and Solution Architectures

A Segment Architecture differs from a Solution Architecture in that the latter often can be described as a discrete and focused business operation or activity and how IT supports that operation. A Solution Architecture, in Figure 10-7 highlighted by the dark-green border, typically applies to a single project or project release.

The Governance Framework (see Chapter 8) includes the roles of Chief Architect, Enterprise Architect, and Domain Architect, but not Segment and Solution Architects.

In the “Architecture Roles and Skills” Series Guide, the term *Segment Architect* is used instead of Domain Architect, and the Solution Architect is mentioned in the context of the Architecture Capability (see Figure 10-4). The Governance Framework does not mention these roles.

Unfortunately, this type of inconsistency occurs periodically in the TOGAF Standard.

During my discussions with The Open Group, we talked about these inconsistencies. They indicated that they are gradually addressing

and changing them. It is possible that the next or an updated version of the framework will be more consistent.

The primary purpose of the Segment Architect role is to focus on a particular segment of the architecture. From a horizontal perspective (based on the four architecture domains), this could be the Business Architecture, Data and Application Architecture, or Technology Architecture domain. If an organization is divided into business domains, chances are that a Segment Architect is responsible for one of these business domains. Often, all four architecture domains fall within such a domain. This is the vertical perspective. In both situations, they are called segments of the architecture.



Further Reading

Additional information about architecture roles and skills is available in the original TOGAF Series Guide “Architecture Skills Framework”, document number G198, and in the updated Series Guide “Architecture Roles and Skills”, document number G249.

10.4. Skills and Competencies

The previous paragraphs have explained the three core architecture roles that the TOGAF Standard prescribes for the proper execution of architecture work:

- Enterprise Architect (with the Chief Architect specialization role)
- Segment Architect (the former Domain Architect role)
- Solution Architect

Each of these roles performs its own part of the architecture work. The Enterprise Architect focuses primarily on the more *strategic aspect* of the profession, while the Segment Architect focuses on detailing and refining *the specific architecture or business domains*. The Solution Architect plays the crucial role of *developing the details* of the proposed solutions. The latter role can therefore be seen as a specialist in a particular subset of the architecture.

Obviously, the proper performance of the various architecture roles requires certain skills and competencies. These are different for each role because of their focus area. The TOGAF Standard has described the skills required for each of the architecture roles listed above in a Series Guide called the “Architecture Skills Framework”. This document provides a set of roles, skills, and proficiency levels for people performing architecture work.

Later, in September 2024 to be exact, The Open Group released an update to the Series Guide. This updated Series Guide is called “Architecture Roles and Skills” (see the *Further Reading* information box).

Skills frameworks provide a view of the competency levels required for specific roles. They define the roles within a work area, the skills required by each role, and the depth of knowledge required to fulfill the role successfully.

Skills frameworks are relatively common for defining the skills required for a consultancy and/or project management assignment, to deliver a specific project or work package. They are also widely used by recruitment and search agencies to match candidates and roles.

To properly perform the work associated with the core architecture roles, it is necessary for architects to have the right set of skills and associated competencies. The TOGAF Standard uses proficiency levels for each skill. In the “Architecture Roles and Skills” Series Guide, the skill categories contain a more detailed overview of the individual skills per category. An overview of the different skill categories [19] is shown in Table 10-1.

Table 10-1. *Skill categories in alphabetical order*

Skill category	Description
Application	Application management, Application Programming Interface (API) patterns, software design principles, etc.
Architecture	Modeling, building block design, high-level design, role definition, Architecture Principle design, high-level migration planning, building block management, systems development and integration methods, Security Architecture, architecture patterns, requirements management, governance, etc.
Business	Enterprise organization knowledge, business cases, business process, strategic planning, etc.
Change	Managing business change, project management methods and tools, etc.
Data	Data analysis, data exchange, and data management.
Generic	Leadership, teamworking, inter-personal skills, etc.
Legal	Data protection laws, contract law, procurement law, fraud, etc.
Technology	Compute, storage, network infrastructure, etc.

The levels of proficiency that can be associated with each of the skills are shown in Table 10-2.

Table 10-2. *Proficiency Levels*

Level number	Name	Description
1	Background	Basic understanding, but reliant on specialists.
2	Aware	Understands and can address issues, supported by specialists.
3	Knowledgeable	Good detail enabling effective decision-making, requiring specialist input only in difficult situations.
4	Expert	Extensive experience, practical, and applied knowledge.

The “Architecture Roles and Skills” Series Guide lists all the skills in the TOGAF Standard and describes these skills in detail. The link to the relevant Series Guide can be found in the *Further Reading* information box at the end of Section 10.3.

Now, in addition to the skills mentioned above, *competencies* also play an important role. Every architect who performs one of the three core architectural roles must have not only the skills, but also the right competencies to perform the skills properly. However, a list of competencies an architect must have is exactly what is missing from the TOGAF Standard.

Competencies differ from skills in that a competency consists of abilities and knowledge in addition to skills. The three together make up a competency. Figure 10-8 illustrates the structure of a competency and the difference between competencies and skills.

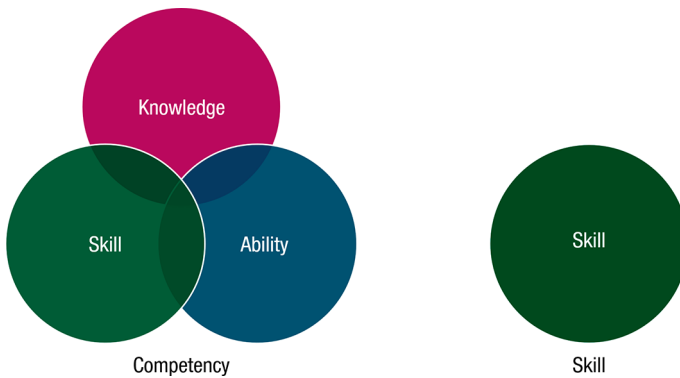


Figure 10-8. *Competencies versus skills*

Having clear and well-articulated competencies in addition to the required skills and associated proficiency levels is essential. Not only from the perspective of the organization, but certainly from the perspective of the architect doing the work.

In fact, I have been in a situation where I worked for an organization that predominantly (if not exclusively) viewed the profession of architecture from an IT perspective. This meant that the skills and competencies that were important to the organization and therefore required of the architect – me in this case – were radically different from the set of competencies I brought to the organization.

My skills and competencies are primarily at the strategic level, being able to translate an organization's strategy into appropriate execution. That requires very different skills compared with implementing IT Architecture.

The organization held on to its view of what it thought an Enterprise Architect should do. As a result, the scope of my work was reduced to that of the IT department. This situation eventually led me to terminate my contract with the organization.

Architecture is an essential component of sound business operations. It provides structure and insight into the coherence of everything an organization does. It is essential that the architect is positioned in the right place in the organization. This position is accompanied by the necessary competencies.

If there is a mismatch between what the architect believes is needed – based on their knowledge, skills, and experience – and how the organization sees the discipline, both parties will be left dissatisfied.

It is therefore highly recommended that a clear and mutually agreed picture of how the work will be done (from which perspective) is established prior to the start of the architecture work, so that the required skills and competencies can be aligned accordingly.

This will prevent unpleasant situations along the way.

The competencies required to properly perform architectural work vary depending on the role of the architect. In general, the following list of competencies applies to the three core architecture roles.

Strategic Thinking and Vision: This competency entails the ability to conceptualize the long-term view of enterprise systems and their alignment with organizational goals. Architects must be able to translate an organization's long-term vision into a concrete strategy. This strategy should then be turned into executable activities or initiatives that should have a place on the Architecture Roadmap.

The competency also requires architects to possess the capability to anticipate future challenges and opportunities to position the organization for success. Architects must have the foresight to help adapt the organization so that the course set can continue. It requires holistic thinking to balance business needs, technology solutions, and operational efficiency.

Strategic Decision-Making: Making informed, strategic decisions under uncertainty or complex scenarios is part of this competency. Architects must be able to make decisions based on informed arguments, even in situations that do not always present a clear picture of the future. These choices will need to be presented to decision-makers. In order for these people to make the right choice, it is important to outline several scenarios so that any risks per choice can be weighed. This means evaluating trade-offs and prioritizing initiatives based on organizational impact and ensuring alignment with organizational objectives while mitigating risks. The Architecture Alternatives and Trade-Offs method (see Chapter 14, Section 14.9) can be used to assist in this process.

Leadership and Influence: Architects need to have the capability to guide and inspire teams toward shared goals. The organization's goals need to be lived by everyone involved. Architects have an important role to play. It is important to build consensus among diverse stakeholders with varying priorities and perspectives. To achieve this, an architect must act as a trusted advisor to executives and decision-makers. The ability to demonstrate leadership contributes to the proper performance of the job.

Change Leadership: A competency closely related to *leadership and influence* is *change leadership*. This competency focuses specifically on driving organization-wide change initiatives with resilience and adaptability (also a separate competency). Part of change leadership is advocating for innovation while addressing resistance and fostering buy-in. Architects should be able to act as a mediator in case of resistance from within the organization. Building support for architectural work is essential to its proper execution. Architects must be sensitive to the culture of an organization. The likelihood of adoption is increased by supporting cultural and structural changes to accommodate new technologies and processes.

Business and Technology Alignment: Architects must be able to bridge the gap between business strategy and technology implementation. Successfully putting this competency into practice requires a solid

understanding of how technology can support or enhance business processes and objectives. It also requires the ability to articulate the value of technology investments in business terms. It is important to continually align the architecture work with organizational goals.

Communication and Collaboration: Architects must demonstrate the ability to communicate complex ideas clearly to both technical and non-technical audiences. They must be able to choose the right tone and context of the conversation at all levels of the organization to get the message across clearly. This helps foster collaboration across multidisciplinary teams and departments. To do this, architects must be able to truly listen and respond constructively to the diverse viewpoints of many stakeholders.

Governance and Compliance: The *governance and compliance* competency focuses on enforcing standards, policies, and guidelines that ensure architectural consistency. Ensuring adherence to regulatory and compliance requirements is a key part of this competency. Another part is monitoring and measuring the effectiveness of architecture initiatives in accordance with the above requirements. This competency addresses the need to establish solid architecture governance. The Architecture Governance Framework described in Chapter 8 can be of assistance.

Adaptability and Resilience: Strategies typically don't change much over the course of a year. However, architects must be able to adapt them to rapidly changing technology landscapes and business environments as needed. Keeping the architecture aligned with business goals and objectives remains critical. Therefore, the ability to maintain composure and thus effectiveness in high-pressure situations is of paramount importance. One of the things that positively contributes to this competency is the ability to continuously learn to stay relevant in evolving domains. This allows architects to stay resilient.

Analytical Thinking and Problem Solving: Another competency architects must possess is the ability to break down complex challenges into manageable components. This means that architects must be able to break down a large problem or challenge into smaller, more manageable pieces. The advantage of this is that each (smaller) part can be considered and addressed as an independent part. Of course, the big picture must always be kept in mind. It is important to synthesize data and insights to identify root causes that can lead to actionable solutions. Architects need to approach problems with creativity and a structured methodology. Architecture frameworks can provide the necessary tools.

Systems Thinking: Since architecture is all about coherence between the various parts and segments of an organization, understanding the interconnectedness of components within an enterprise ecosystem is an absolute must. Architects should be able to design scalable and flexible systems and solutions that work cohesively across departments, with an eye toward the broader fit within the organization. They should focus on the sustainability and adaptability of solutions over time, which is related to the *adaptability and resilience* competency.

Table 10-3 provides an overview of the required competencies detailed above and maps them to the three core architecture roles.

Table 10-3. *Required architecture competencies per core architecture role*

Competency	Description	Core Architecture Role
Strategic thinking and vision	Ability to conceptualize the long-term view of enterprise systems and their alignment with organizational goals.	Enterprise Architect
	Capability to anticipate future challenges and opportunities to position the organization for success.	Enterprise Architect, Segment Architect
	Holistic thinking to balance business needs, technology solutions, and operational efficiency.	Enterprise Architect, Segment Architect
Strategic decision-making	Making informed, strategic decisions under uncertainty or complex scenarios.	Enterprise Architect
	Evaluating trade-offs and prioritizing initiatives based on organizational impact.	Enterprise Architect, Segment Architect
	Ensuring alignment with organizational objectives while mitigating risks.	Enterprise Architect, Segment Architect
Leadership and influence	Capability to guide and inspire teams toward shared goals.	Enterprise Architect
	Building consensus among diverse stakeholders with varying priorities and perspectives.	Enterprise Architect
	Acting as a trusted advisor to executives and decision-makers.	Enterprise Architect

(continued)

Table 10-3. *(continued)*

Competency	Description	Core Architecture Role
Change leadership	Driving organization-wide change initiatives with resilience and adaptability.	Enterprise Architect
	Advocating for innovation while addressing resistance and fostering buy-in.	Enterprise Architect, Segment Architect
	Supporting cultural and structural changes to adopt new technologies and processes.	Enterprise Architect, Segment Architect
Business and technology alignment	Competence in bridging the gap between business strategies and technology implementations.	Enterprise Architect, Segment Architect
	Strong understanding of how business processes and objectives can be supported or enhanced by technology.	Enterprise Architect, Segment Architect
	Ability to articulate the value of technology investments in business terms.	Segment Architect
Communication and collaboration	Proficiency in conveying complex ideas clearly to both technical and non-technical audiences.	Segment Architect, Solution Architect
	Fostering collaboration across multidisciplinary teams and departments.	Segment Architect, Solution Architect

(continued)

Table 10-3. *(continued)*

Competency	Description	Core Architecture Role
Governance and compliance	Listening and responding constructively to diverse viewpoints.	Enterprise Architect, Segment Architect, Solution Architect
	Enforcing standards, policies, and guidelines that ensure consistency in architecture.	Enterprise Architect, Segment Architect, Solution Architect
	Ensuring adherence to regulatory and compliance requirements.	Enterprise Architect, Segment Architect
Adaptability and resilience	Monitoring and measuring the effectiveness of architecture initiatives.	Segment Architect, Solution Architect
	Adjusting to rapidly changing technology landscapes and business environments.	Segment Architect, Solution Architect
	Maintaining composure and effectiveness during high-pressure situations.	Enterprise Architect, Segment Architect, Solution Architect
	Continuously learning to stay relevant in evolving domains.	Enterprise Architect, Segment Architect, Solution Architect
Analytical thinking and problem solving	Breaking down complex challenges into manageable components.	Enterprise Architect, Segment Architect, Solution Architect

(continued)

Table 10-3. *(continued)*

Competency	Description	Core Architecture Role
Systems thinking	Synthesizing data and insights to identify root causes and actionable solutions.	Solution Architect
	Approaching problems with creativity and a structured methodology.	Segment Architect, Solution Architect
	Understanding the interconnectedness of components within an enterprise ecosystem.	Segment Architect, Solution Architect
	Ability to design scalable and flexible systems that work cohesively across departments.	Segment Architect, Solution Architect
	Focus on sustainability and adaptability of solutions over time.	Segment Architect, Solution Architect

The competencies listed in Table 10-3 collectively ensure that architects not only design effective systems but also drive organizational transformation and deliver business value.

Each of the competencies listed in Table 10-3 corresponds to a skill listed in the “Architecture Roles and Skills” Series Guide. Figure 10-9 shows the relationships between competencies and skills.

Skill	Description	Architecture Competency									
		Strategic thinking and vision	Strategic decision-making	Leadership and influence	Change leadership	Business and technology alignment	Communication and collaboration	Governance and compliance	Adaptability and resilience	Analytical thinking and problem solving	Systems thinking
Application	Application management, Application Programming Interface (API) patterns, software design principles, etc.					X		X	X	X	X
Architecture	Modeling, building block design and management, high-level design, role definition, Architecture Principle design, high-level migration planning, systems development and integration methods, security architecture, architecture patterns, requirements management, governance, etc.		X	X	X	X	X	X		X	
Business	Enterprise organization knowledge, business cases, business process, strategic planning, etc.	X	X	X	X	X	X	X			
Change	Managing business change, project management methods, and tools, etc.		X	X	X	X	X	X	X		
Data	Data analysis, data exchange, and data management.							X	X	X	X
Generic	Leadership, teamworking, inter-personal skills, etc.		X	X		X	X				
Legal	Data protection laws, contract law, procurement law, fraud, etc.		X					X	X	X	
Technology	Compute, storage, and network infrastructure, etc.					X		X	X	X	X

Figure 10-9. Architecture competencies mapped to skills

At this time, the TOGAF Standard does not include an overview or listing of architecture competencies. Hopefully, this will be added in one of the future updates of the framework.



Further Reading

The SFIA 9 skills directory provides an overview of skills and competencies for Segment and Solution Architects:

<https://sfia-online.org/en/sfia-9/all-skills-a-z>

10.5. Summary

This chapter has highlighted the various architecture roles that exist according to the TOGAF Standard and are needed to properly implement the Architecture Capability. The TOGAF Standard introduces three primary views of the architecture profession: enterprise, segment, and solution. These three primary viewpoints correspond to the three core architecture roles:

- Enterprise Architect (with the Chief Architect specialization role)
- Segment Architect (the former Domain Architect role)
- Solution Architect

Each of the three roles performs its own part of the overall architecture work.

The *Chief Architect* is a special case of an Enterprise Architect and is responsible for integrating the Enterprise Architecture view into the wider strategic and operational context of the organization.

Enterprise Architects lead major initiatives and hold a seat at the table with senior decision-makers. They are accountable for comprehensive

business service delivery through information systems. They demonstrate advanced expertise in both technical and business arenas.

The role of the *Segment Architect* is all about defining, shaping, and governing the systems, subsystems, and components that bring products and services to life. Segment Architects build roadmaps, create blueprints for the future of their segments, and ensure every technical and business option is thoroughly evaluated and effectively utilized.

Solution Architects are seasoned professionals who drive the delivery of new or evolved capabilities across the organization and its marketplace. Their role is to craft and guide the solutions that align with both immediate requirements and broader organizational goals, achieving defined service levels within ambitious timelines. They break down these solutions into systems, subsystems, and components that interact seamlessly and then integrate them into a cohesive roadmap, transforming complex ideas into actionable steps through targeted Transition Architectures.

There are several ways to look at an organization from an architectural perspective. The first way is from a horizontal perspective. This involves dividing the organization into the four well-known architecture domains (Business, Data, Application, and Technology). The second perspective is called the vertical perspective. This looks at the services of an organization and shows them in vertical segments. A vertical segment usually consists of one or more (or even all four) architecture domains.

Skills and competencies are required to properly perform the aforementioned architecture roles. The TOGAF Standard only describes the required skills and the required levels of proficiency. Because skills alone are not sufficient to perform architecture work or to fulfill the role of an architect, this book includes a list of ten competencies. These competencies are mapped to the skill set created by the TOGAF Standard.